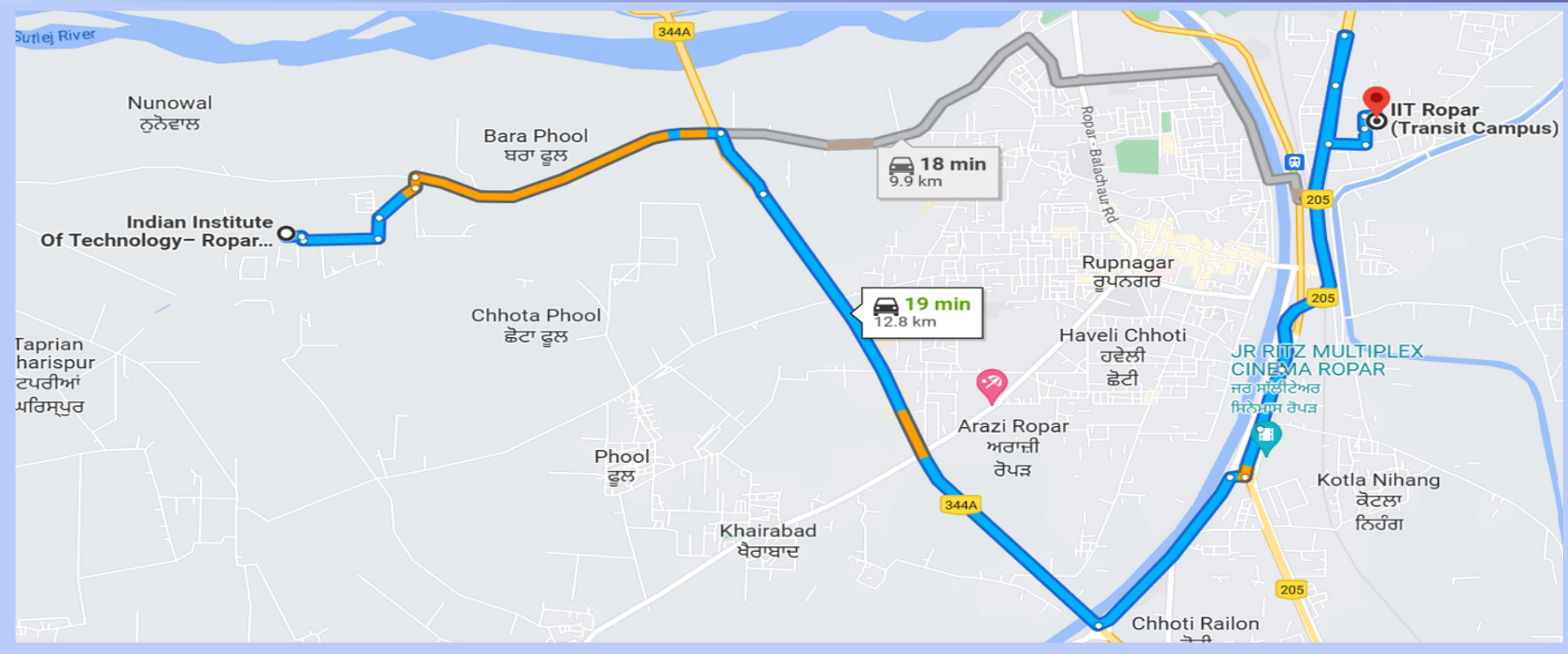


Constrain Path Optimization on Time-Dependent Road Networks

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Nuanced Routing Queries



Google Map

With increasing urban mobility big data, one can ask nuanced queries such as:

- Determine a path that maximizes safety or navigability while constraining the total length of the path to be at most 10 minutes longer than the fastest path.
- Find a route with the most scenic views, with a constraint that the total length of the path does not exceed 15 minutes more than the fastest route.

Problem Definition

Input:

- Input directed graph $G(V, E)$ is a time-dependent graph. Each edge property (score m , travel time c) is time-dependent.
- Source S and Destination D .
- Departure time from S as t_{dep}^S , Budget constrain Θ .

Output: A path $\mathcal{P}$ from S to D , which maximizes total m and total $c \leq \Theta$.

Contributions

- Introduce *SCOPE*, to solve the TD-CPO problem. *SCOPE* uses a recursive approach to explore a broader search space, yielding higher-quality solutions.
- *SCOPE* incorporates temporal pruning, significantly reducing the search space and improving efficiency.
- *SCOPE* utilizes a depth-first exploration strategy, allowing us to put an upper limit on the allowed execution time. *SCOPE* also uses a parallel execution strategy to get better solutions within the user-defined time limit.

References

- [1] Ying Lu et al.: *Scenic routes now: Efficiently solving the time-dependent arc orienteering problem*. The ACM Conference on Information and Knowledge Management CIKM (2017)
- [2] Liping Gao et al.: *A Memetic Algorithm for Finding the Two-fold Time-dependent Most Beautiful Driving Routes*. IEEE Wireless Communications and Networking Conference WCNC (2019)
- [3] Chao Chen et al.: *Enjoy the most beautiful scene now: a memetic algorithm to solve two-fold time-dependent arc orienteering problem*. Front. of Comp. Sc. (2019)
- [4] Ye Yuan et al.: *Weight-Constrained Route Planning Over Time-Dependent Graphs*. IEEE International Conference on Data Engineering ICDE (2019)
- [5] Ye Yuan et al.: *Constrained Shortest Path Query in a Large Time-Dependent Graph*. Proc. VLDB Endow. (2019)

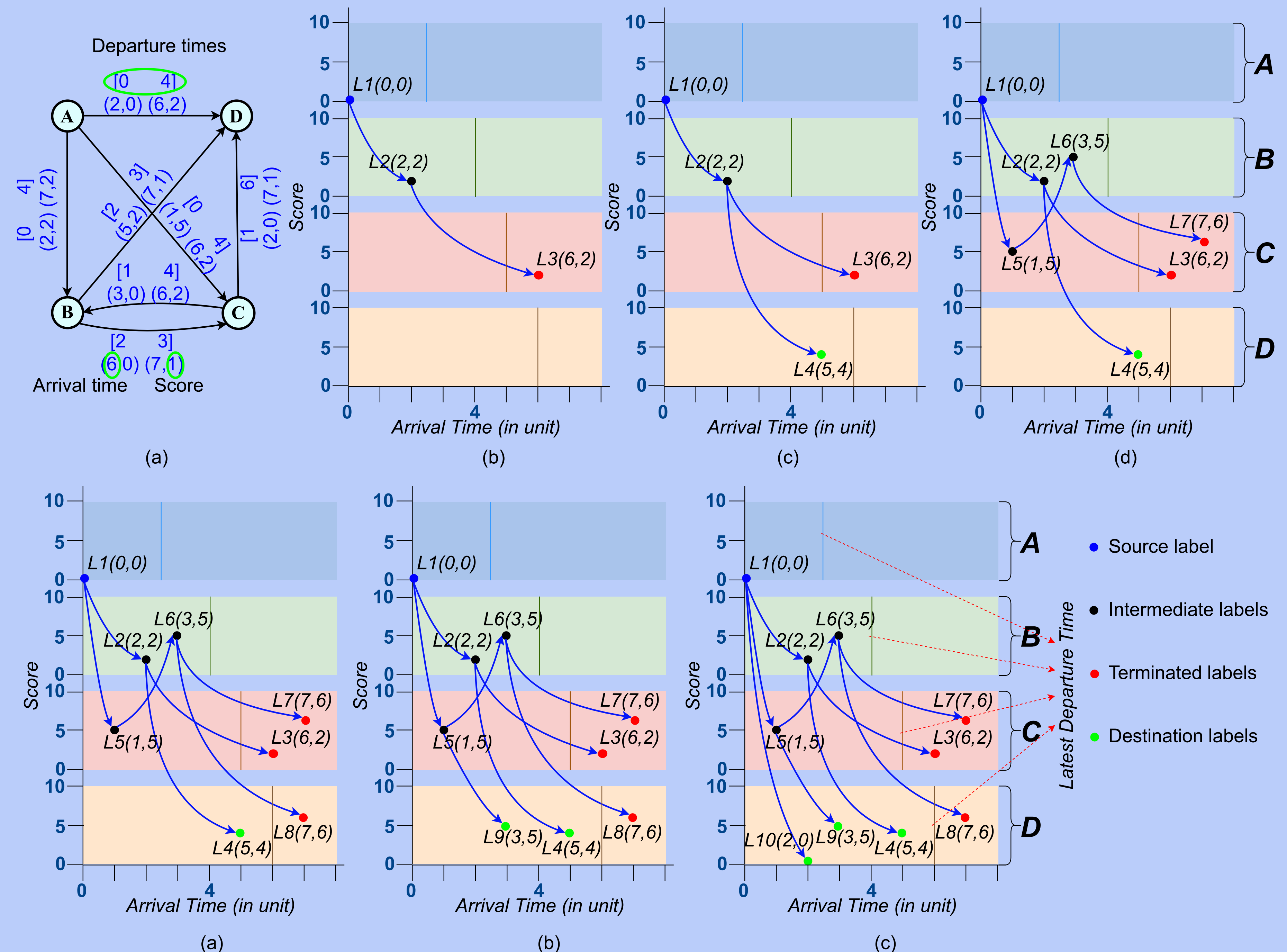
Acknowledgements



Limitations of State-of-the-arts

- Relying on heuristic based selection of a single seed path (as done in [1]) often leading to low solution quality.
- Handling multiple seed paths (as done in [2, 3]) is computationally intensive and does not scale efficiently for large networks.
- Adapting time-dependent CSP algorithms (e.g., [5, 4]) to solve an instance of the TD-CPO problem does not yield a one-to-one mapping for ensuring high-quality solutions [5, 4].

Working of *SCOPE*

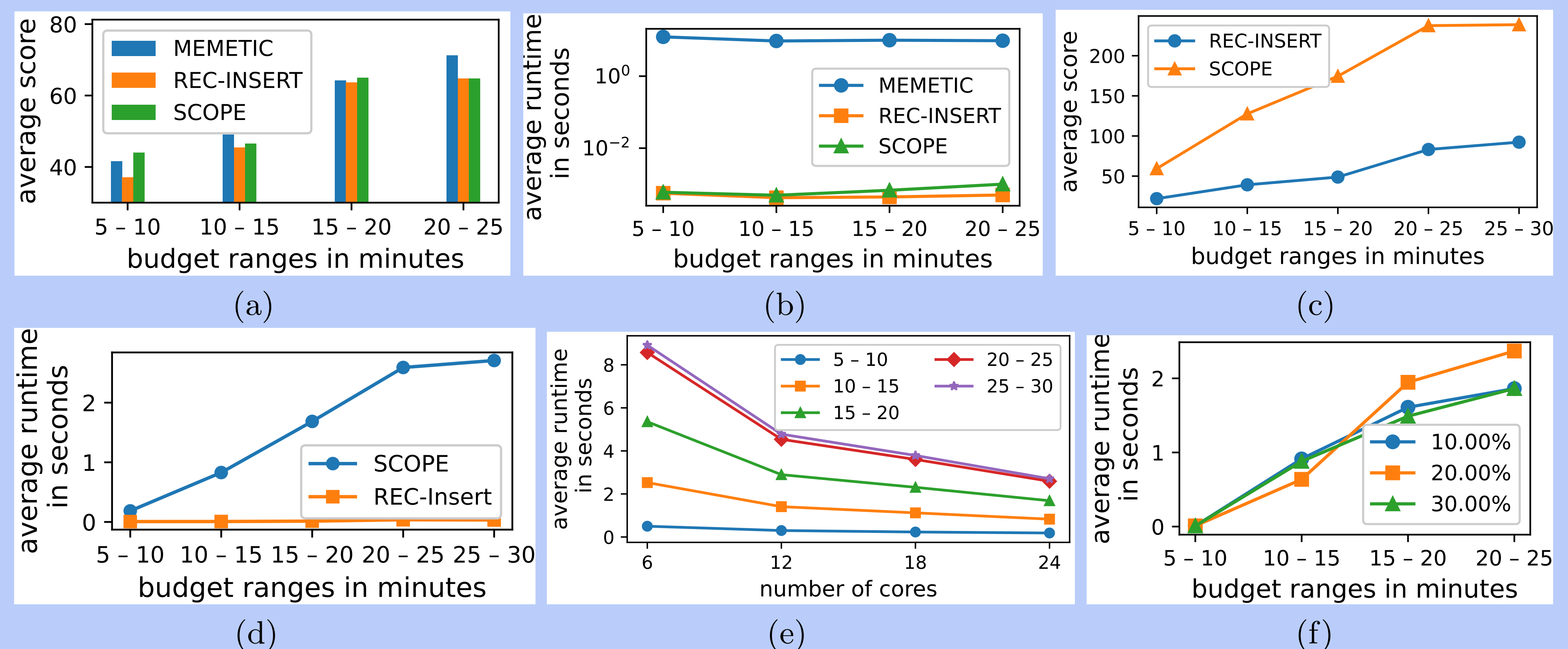


Example query with source A , destination D , departing A at $t=0$ and budget constraint 6.

Experimental Analysis

- **Dataset:** (1) Oldenburg (6k nodes, 7k edges), (2) Moscow (685k nodes, 1.8M edges).
- **Candidate algorithms:** (1) REC-INSERT[1], (2) MEMETIC[3].
- **Setup:** Intel Xeon Gold 6258R server with 28 cores, 32GB of RAM, a base frequency of 2.7GHz, and a maximum boost frequency of 4GHz.
- **Query set:** Four query bucket with different budget ranges with 100 queries each.
- **Evaluation metrics:** (1) average runtime and (2) average score gain in each query bucket.

Results:



(a) & (b) Score & Runtime comparison in Oldenburg network, (c) & (d) Score & Runtime comparison in Moscow network, (e) & (f) Runtime comparison with varying number of cores & Runtime comparison with varying density of edges with positive score in Moscow network. For all experiments, overhead is fixed to 50% over the fastest path cost.